

Pico Gym Exclusive - Bit-O-Asm-2

Bit-O-Asm-2

Reverse Engineering

Medium

100 pts

by LT 'syreal' Jones

Can you figure out what is in the `eax` register? Put your answer in the picoCTF flag format: `picoCTF{n}` where `n` is the contents of the `eax` register in the decimal number base. If the answer was `0x11` your flag would be `picoCTF{17}`.

In this challenge, we're asked to look at an assembly dump for a program and figure out what's in the `eax` register at the end of the program

Pico Gym Exclusive – Bit-O-Asm-2 x86 Assembly Instructions

But to understand the contents of the disassembly dump file, we'll need to understand a bit about the x86 assembly language



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Assembly languages are the programming languages closest to the pure machine code that computer processors execute



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The EAX Memory Register

```
mov     DWORD PTR [rbp-0x4],0x9fe1a  
mov     eax,DWORD PTR [rbp-0x4]
```

The goal of the challenge is to understand what value is in the EAX register, but what is it?

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The EAX Memory Register

```
mov     DWORD PTR [rbp-0x4],0x9fe1a  
mov     eax,DWORD PTR [rbp-0x4]
```

In x86 assembly, EAX is a memory register, a location where data is stored. The CPU has many memory registers, and EAX is a very commonly-used one

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The MOV Instruction

```
MOV <location affected>, <value>
```

MOV is an assembly instruction which moves data into memory locations

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The MOV Instruction

```
mov     DWORD PTR [rbp-0x4],0x9fe1a  
mov     eax,DWORD PTR [rbp-0x4]
```

So, in these instructions, MOV will move the value 0x9fe1a (decimal value 654874) into the DWORD PTR location...

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The MOV Instruction

```
mov     DWORD PTR [rbp-0x4],0x9fe1a  
mov     eax,DWORD PTR [rbp-0x4]
```

Then, another MOV instruction will move the contents of the DWORD PTR location to the EAX register